

Day 10 Soc Internship

Threat intelligence & OSINT Basics

Assignment:

- 1 Explain threat intelligence types
- 2 Describe how to investigate an IP
- 3 Give one real-world example

(1)

Threat intelligence is information used to understand cyber threats, attackers, and attack methods. It helps security teams detect, prevent, and respond to attacks.

1. Strategic Threat Intelligence

Strategic intelligence is high-level information about cyber threat trends. It is mainly used by managers, executives, and decision-makers.

Example:

A report showing that ransomware attacks are increasing against healthcare and financial organizations.

2. Tactical Threat Intelligence

Tactical intelligence focuses on attacker techniques, tactics, and procedures, also called **TTPs**. It helps security teams understand how attackers perform attacks.

Example:

Attackers using phishing emails, PowerShell scripts, credential dumping, or lateral movement techniques.

3. Operational Threat Intelligence

Operational intelligence gives details about specific attacks, threat groups, or campaigns. It helps security teams prepare for active or upcoming threats.

Example:

A known threat group targeting Microsoft Exchange servers using a specific exploit campaign.

4. Technical Threat Intelligence

Technical intelligence includes specific indicators of compromise, also called **IOCs**. These are used directly in security tools like SIEM, firewall, EDR, and IDS.

Examples:

Malicious IP addresses, suspicious domains, file hashes, URLs, email addresses, and malware signature

(2)

1. Identify the IP Address

First, check the IP address from the alert or log.

Find out whether the IP is:

Internal IP

External IP

Public IP

Private IP

Private IP ranges are usually internal:

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

If the IP is public, it may belong to an external attacker, cloud service, VPN, proxy, or legitimate vendor.

2. Check the Log Details

Review the full log where the IP appeared.

Time of activity

Source IP

Destination IP

Username

Device name

Action allowed or blocked

Port number

Protocol

Number of attempts

Success or failure

3. Check IP Reputation

Search the IP in threat intelligence sources.

VirusTotal ,AbuseIPDB, AlienVault OTX. Cisco Talos, IBM X-Force. GreyNoise

Look for:

Malicious score, Reported abuse, Malware activity, Phishing activity, Botnet activity, Scanner activity
Country and ASN, First seen / last seen

4. Check Geo-location and Ownership

Find out where the IP belongs.

Country

ISP / Organization

ASN
Hosting provider
Cloud provider
VPN / Proxy / Tor

5. Check Internal Logs

Search the IP in your SIEM.

Check whether the same IP appears in:

Firewall logs

VPN logs

Proxy logs

Authentication logs

EDR alerts

DNS logs

Web server logs

Email gateway logs

6. Check the Type of Activity

Activity	Possible Meaning
Many failed logins	Brute-force attack
Connection to many ports	Port scanning
Large data upload	Data exfiltration
Access to malicious domain	Malware communication
Login from unusual country	Account compromise
Successful login after failures	Possible brute-force success

7. Check If the IP Is Allowed or Known

Before blocking, confirm whether the IP is legitimate.

Is it a company vendor?

Is it a cloud service?

Is it part of VPN?

Is it used by employees?

Is it in an allowlist?

Is it linked to a business application?

8. Check Related Users and Devices

Find which user or device communicated with the IP.

9. Decide Severity

Classify the IP activity based on evidence.

10. Take Action

Based on severity, take proper action.

The IP address should be investigated by checking its reputation, ownership, geolocation, related logs, affected users, and activity pattern. If the IP shows malicious reputation, repeated failed logins, scanning behavior, malware communication, or successful login activity, it should be treated as suspicious. The analyst should block the IP if confirmed malicious, investigate affected systems, and escalate the case for deeper analysis.

(3)

Real-World Example: Suspicious IP Investigation

Scenario

A SOC analyst receives an alert from the SIEM:

Alert Name: Multiple Failed Login Attempts

Source IP: 185.220.101.45

Destination Host: VPN-Gateway

Username: admin

Event Count: 75 failed logins in 5 minutes

Action: Failed Login

Investigation Steps

1. Check the IP Address

The analyst first checks whether the IP is internal or external.

Source IP: 185.220.101.45

Type: Public IP

2. Check IP Reputation

The analyst checks the IP in threat intelligence platforms like:

VirusTotal

AbuseIPDB

AlienVault OTX

Cisco Talos

3. Check SIEM Logs

The analyst searches the same IP in SIEM.

src_ip="185.220.101.45"

Findings:

75 failed VPN login attempts

Targeted usernames: admin, test, user1

No successful login found

Same IP tried login on multiple accounts

4. Check Affected Accounts

The analyst checks whether any account was successfully accessed.

admin - failed only

test - failed only

user1 - failed only

5. Check Geo-location and ASN

The analyst checks IP ownership.

Country: Foreign location

ISP: Hosting/VPN provider

ASN: Data center network

Analyst Conclusion

The IP 185 . 220 . 101 . 45 is suspicious because it generated many failed login attempts against the VPN gateway within a short time. It also targeted multiple usernames and has a bad reputation in threat intelligence sources. No successful login was found, so there is no confirmed account compromise, but the activity matches brute-force behavior.